

CANDIDATE NAME _____

CG _____



SERANGOON JUNIOR COLLEGE
JC2 PRELIM PRACTICAL EXAMINATION 2017

H2 BIOLOGY

9744
2.5 hours

INSTRUCTIONS TO CANDIDATES

Write your name, CG and index number in the spaces at the top of this page.

Write in dark blue or black pen.

The use of an approved scientific calculator is expected, where appropriate. You may lose marks if you do not show your working or if you do not use appropriate units.

Answer **all** questions in the spaces provided on the Question Paper.

FOR EXAMINER'S USE	
Question	
1	/13
2	/14
3	/14
4	/14
TOTAL	/55

INFORMATION FOR CANDIDATES

The intended number of marks is given in brackets [] at the end of each question or part question.

This question paper consists of 14 printed pages.

Question 1

You are provided with a quantity of vitamin C solution and a dye called DCPIP.

You are also provided with three test-tubes containing respectively lemon juice, orange juice, grapefruit juice, and labelled as such. These juices contain natural vitamin C and the dye DCPIP can be used to determine the concentration of this vitamin in the juices.

Apparatus: 6 test-tubes and a test-tube rack
4 plastic teat pipettes
Plastic ruler

Material: 30 cm³ DCPIP solution, labelled '**DCPIP**'
50 cm³ vitamin C solution, labelled '**Vitamin C solution**'
50 cm³ lemon juice
50 cm³ orange juice
50 cm³ grapefruit juice

Proceed as follows:

1. Into a clean test-tube, transfer a quantity of the dye DCPIP to a depth of 0.5 cm. Take note its colour.
2. Fill a teat pipette with vitamin C solution. Add one drop of vitamin C solution to the DCPIP solution in the test-tube and shake gently. Continue to add the drops, counting the number of drops which are needed to bring about a colour change. Shake gently after each drop, refilling the pipette if necessary.
3. Record the initial colour of DCPIP (from step 1) and the first colour change after vitamin C is added as well as the number of drops counted to bring about this colour change in a suitable table after step 9.
4. After the first colour change, continue adding drops of vitamin C and counting the drops until the DCPIP solution becomes colourless/or consistent pale yellow. (Ignore any coloured granules that might form.). Record the number of drops counted in the same table from step 3.
5. Repeat steps 1 to 4 adequately to obtain enough data for analysis, cleaning all apparatus before use.
6. Place the DCPIP solution into each of three clean test-tubes to a depth of 0.5 cm. (The amount of DCPIP solution must be exactly the same in each of the tubes). Label the tubes A, B and C.
7. Fill a clean teat pipette with lemon juice and drop by drop add this to the contents of tube **A**, shaking the tube gently after each drop. Count the number of drops needed to turn the DCPIP solution colourless. Repeat this step adequately to obtain enough data for analysis.
8. Repeat the step 7 with orange juice and grapefruit juice, using a clean pipette each time to add the juice to the DCPIP solution in tubes **B** and **C** respectively.

9. Record the results for the three juices and the earlier results for DCPIP and Vitamin C in an appropriate table. [5]

10. What conclusions can you draw from your results? [2]

.....

.....

.....

.....

11. Comment on the main source(s) of error and the limitations of the measurements or experimental procedure. [3]

.....

.....

.....

.....

.....

.....

.....

.....

12. What improvements could you make to the experimental procedures to overcome these sources of error? [3]

.....

.....

.....

.....

.....

.....

.....

.....

[Total: 13 marks]

Question 2

You are provided with a leaf labelled Q. In this question you will be required to investigate the number of stomata present on the lower surface of leaf Q.

Proceed as follows:

1. Use nail varnish to cover a small area of the lower epidermis of leaf Q. Apply a thin layer over an area about 1cm^2 . Avoid any large veins that may be present.
2. Repeat this process for three different areas of the lower epidermis.
3. Allow the nail varnish about 20 minutes to dry. (Proceed to question 3 while the leaf is drying)
4. After 20 minutes, use a razor blade or a fine scalpel to lift one edge of the layer of nail varnish. Use forceps to then gently peel a layer of nail varnish off the leaf.
5. Transfer this layer of nail varnish to a slide and cover with a cover slip.
6. Examine the slide using a microscope and count the number of stomata you can observe in the field of view.
7. State which objective you chose to make this stomatal count and explain your choice. [1]

.....

.....

.....

8. Repeat the counting of stomata in the field of view for a second piece of peeled nail varnish. Calculate the mean number of stomata per field of view in the space below. [2]

9. Make a high power drawing of three adjacent stomata from either of your stomatal peels. [3]

10. Using the eyepiece graticule in your microscope and the provided stage micrometer, find the actual length, in μm , of one of the guard cells that you have drawn. [3]

A researcher obtained leaves from two different plants, Plant A and Plant B. From the same forest. He found that the number of stomata on the underside of the both leaves differed. He ensured that he calculated the number of stomata based on per unit area and ensured he had sufficient replicates. However, the numbers still did not match. The mean density of stomata per 1cm^2 for leaves A and B were as follows:

Leaf A – 234

Leaf B – 297

11. While the researcher expected a difference in the number of stomata, he could not be sure if the difference seen was significant. Suggest a statistical test he could use to confirm if the difference was significant. [1]

.....

12. When carried out this test, the probability value he obtained was less than 0.05. Comment on what these results show and suggest an explanation for the pattern seen in Leaves A and B collected by the researcher. [4]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

[Total: 14 marks]

Question 3

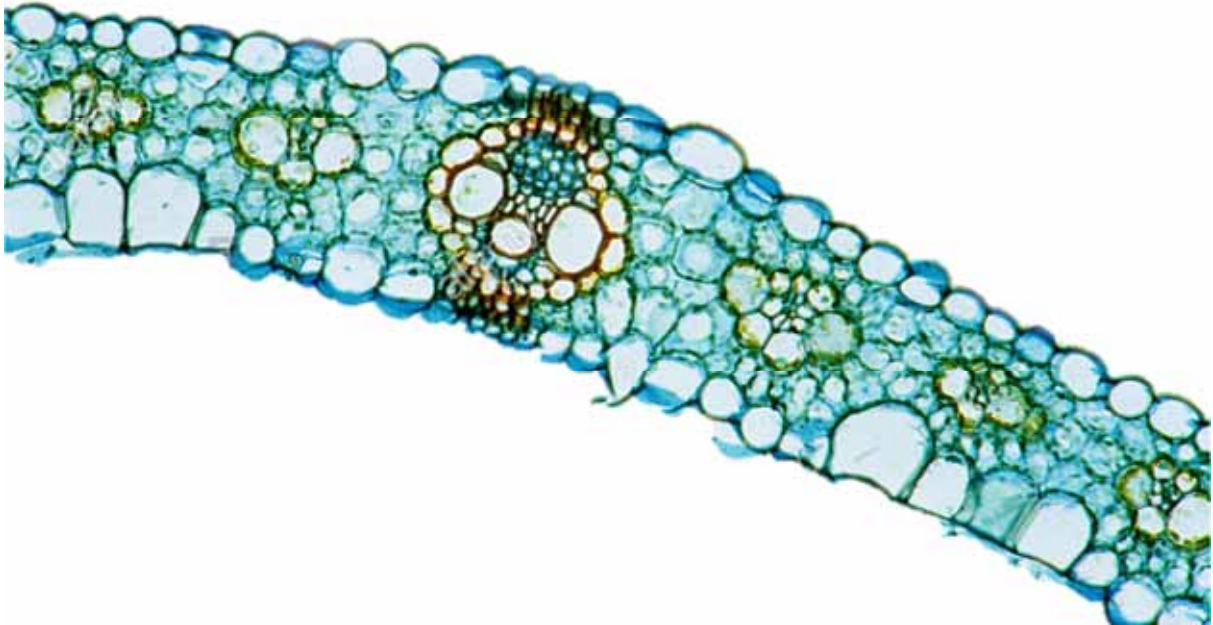
Slide S1 is a tranverse section from a leaf of the *Amophilia* plant.

(a) In the space below, draw a labelled plan drawing of this leaf. [6]

- (b) Using the provided **slide graticule**, measure the diameter of the leaf. You may assume that the slide graticule can be used as you would use a ruler. [1]

Diameter of leaf: _____

The following **Figure 2.1** is a microscope image of a section of a typical monocot leaf.



From alamy.com

Figure 2.1

- (c) Given that the magnification of Figure 2.1 is 50X, calculate the actual width of the specimen. [2]

(d) Identify three differences between a typical monocot leaf and the *Amophilia* leaf. [3]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(e) Suggest reasons for the differences you have identified. [2]

.....

.....

.....

.....

.....

.....

[Total: 14 marks]

Question 4

The Tasmanian Tiger or Thylacine (*Thylacinus cynocephalus*), the world's largest carnivorous marsupial, was once common throughout Australia and Papua New Guinea. The Thylacine resembled a large, short-haired dog with a stiff tail which smoothly extended from the body in a way similar to that of a kangaroo. The female Thylacine had a pouch with four teats, but unlike many other marsupials, the pouch opened to the rear of its body.

An example of convergent evolution, the Thylacine showed many similarities to the members of the Canidae (dog) family of the Northern Hemisphere: sharp teeth, powerful jaws, raised heels and the same general body form.

Due to human activities, the Thylacine was hunted to extinction by early 1930. A Thylacine specimen with soft tissue remaining is found in the Australian Museum in Sydney.

Imagine that you are a researcher in the Australian Rare Fauna Research Association. Recently it was reported that locals near Mount Carstensz in Western New Guinea had sighted creatures that resemble Thylacines. Some members of the Association believe that the creatures sighted may be descendents of the Thylacine, while other members believe that the creatures may be a new species. If the former were true, then the Thylacine is not extinct and conservation effects may revive the species.

Plan an investigation to investigate if these creatures found in Western New Guinea are descendents of the Thylacine that was thought to be extinct.

Your planning must be based on the assumption that you have been provided with the following equipment and materials.

- Tissue sample from the museum specimen and from the Western New Guinea creatures under investigation
- Pestle and mortar
- DNA extraction buffer solution
- Microcentrifuge tubes
- Centrifuge
- Restriction enzymes
- Agarose or polyacrylamide gel plate
- Suitable source of electric current
- Radioactive probe
- Nitrocellulose membrane
- Autoradiography equipment

Your plan should have a clear and helpful structure to include:

- An explanation of the theory to support your practical procedure
- A description of the method used including scientific reasoning behind the method
- The type of data generated by the experiment
- How the results will be analysed including how the origin of the organism can be determined

[Total: 14]

[illegible]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

End of Paper